INCH-POUND
MIL-DTL-16377/14B(SH)
30 June 2014
SUPERSEDING
MIL-F-16377/14A
18 March 1982

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING;T5 FLUORESCENT AND SSL EQUIVALANT, GENERAL LIGHTING, 8 WATTS, 120 VOLTS, 60 HERTZ, WATERTIGHT, SYMBOLS 79.1, 79.2, 80.1, 80.2, 79.1SSL, 79.2SSL, 80.1SSL, AND 80.2SSL

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

All dimensions are in inches. For tolerances other than specified herein, see 3.5.2 of MIL-DTL-16377.

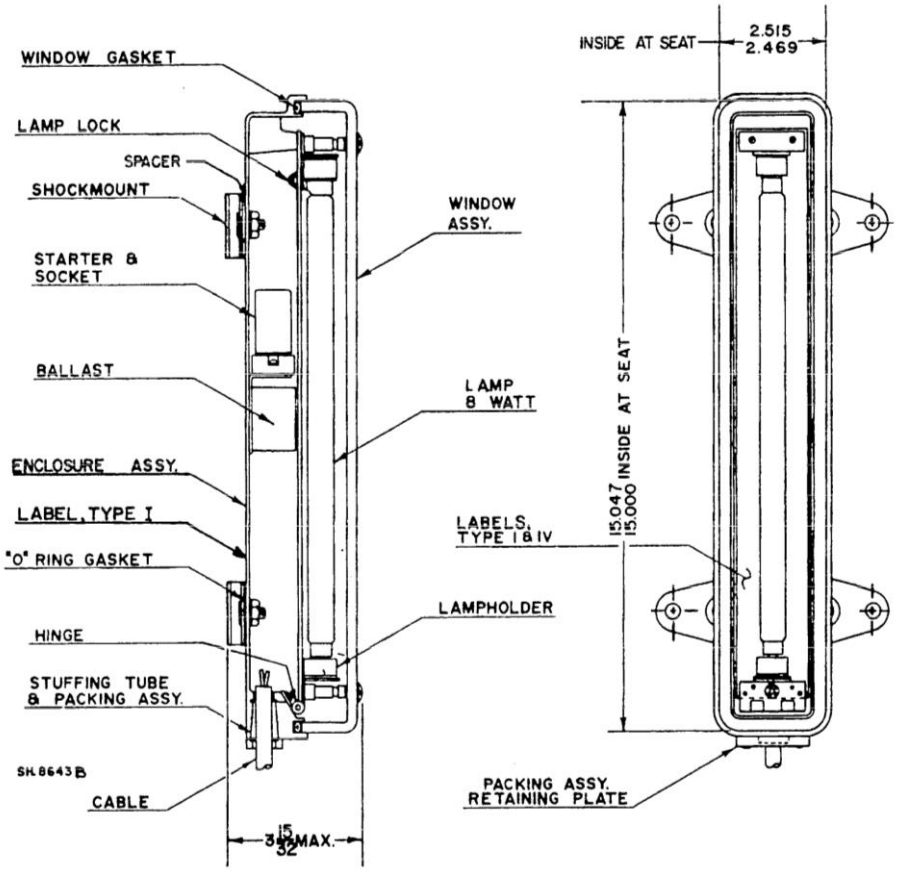


FIGURE 1. Typical view of symbols 79.1 and 80.1.

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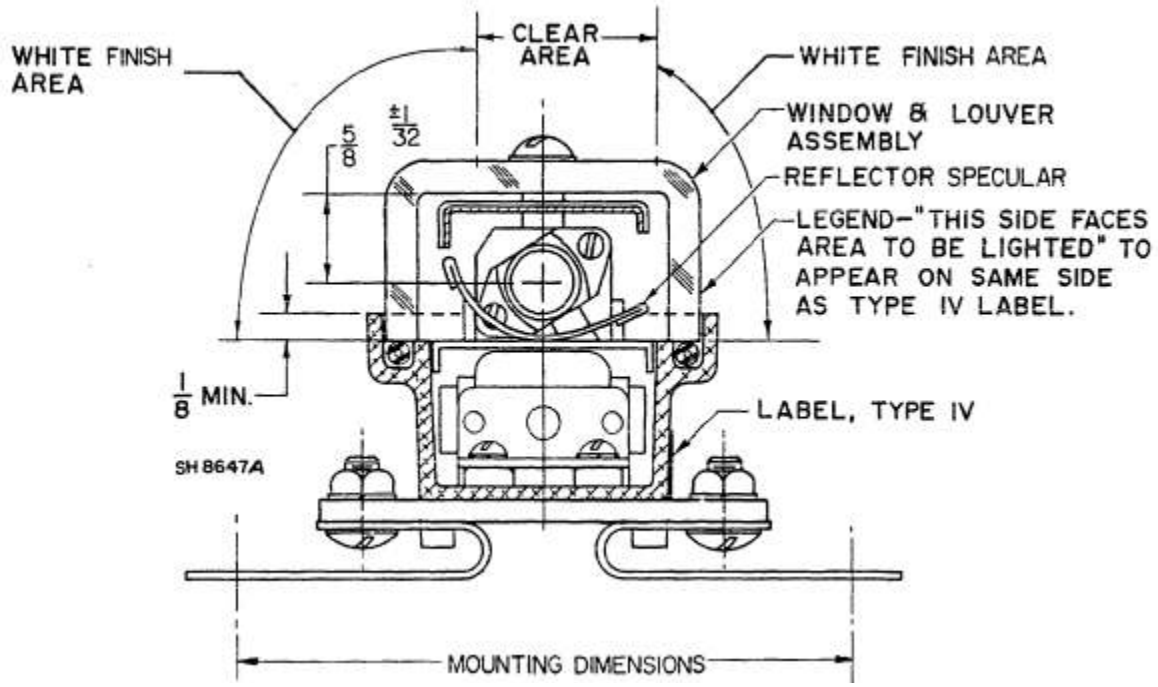


FIGURE 2. Typical section for symbol 79.1.

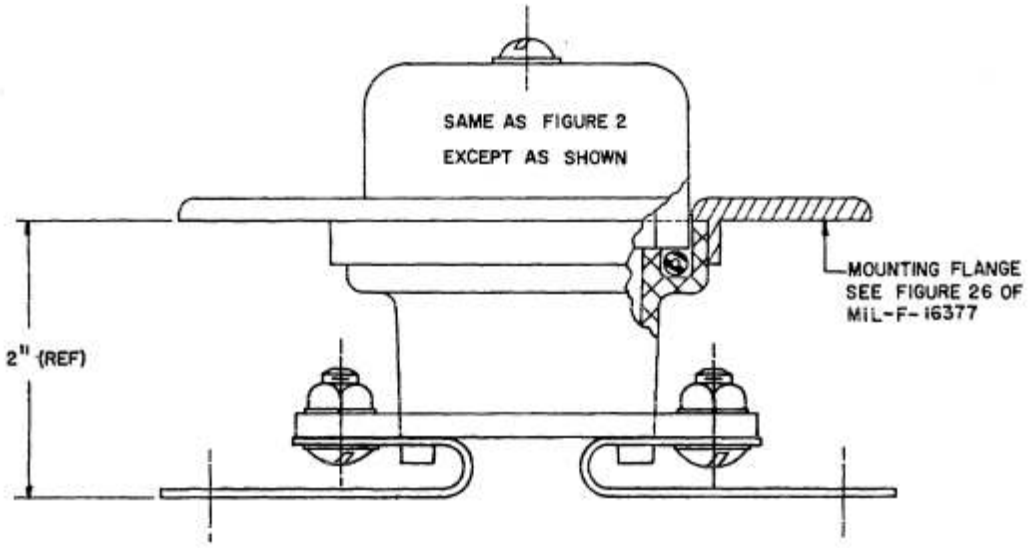


FIGURE 3. Typical section for symbol 79.2.

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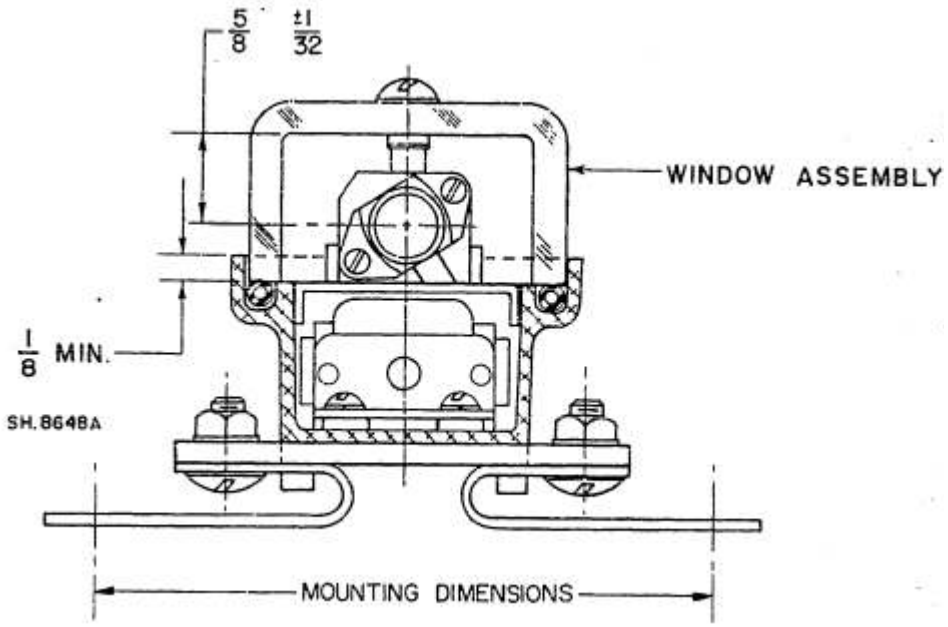


FIGURE 4. Typical section for symbol 80.1.

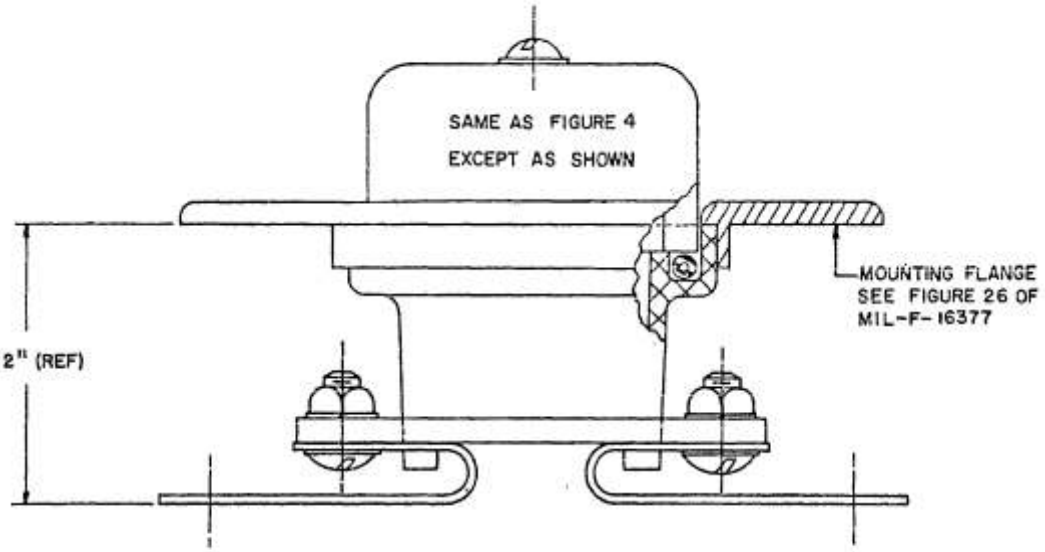
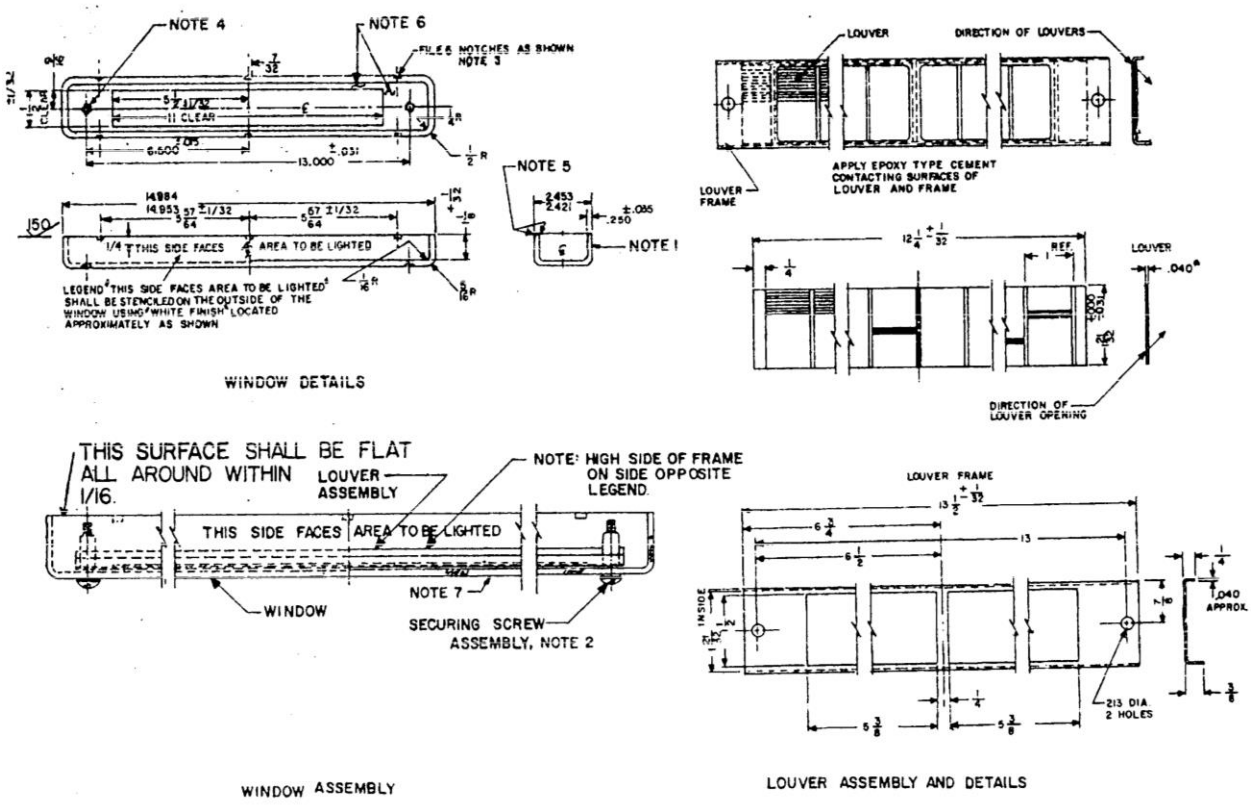


FIGURE 5. Typical section for symbol 80.2.

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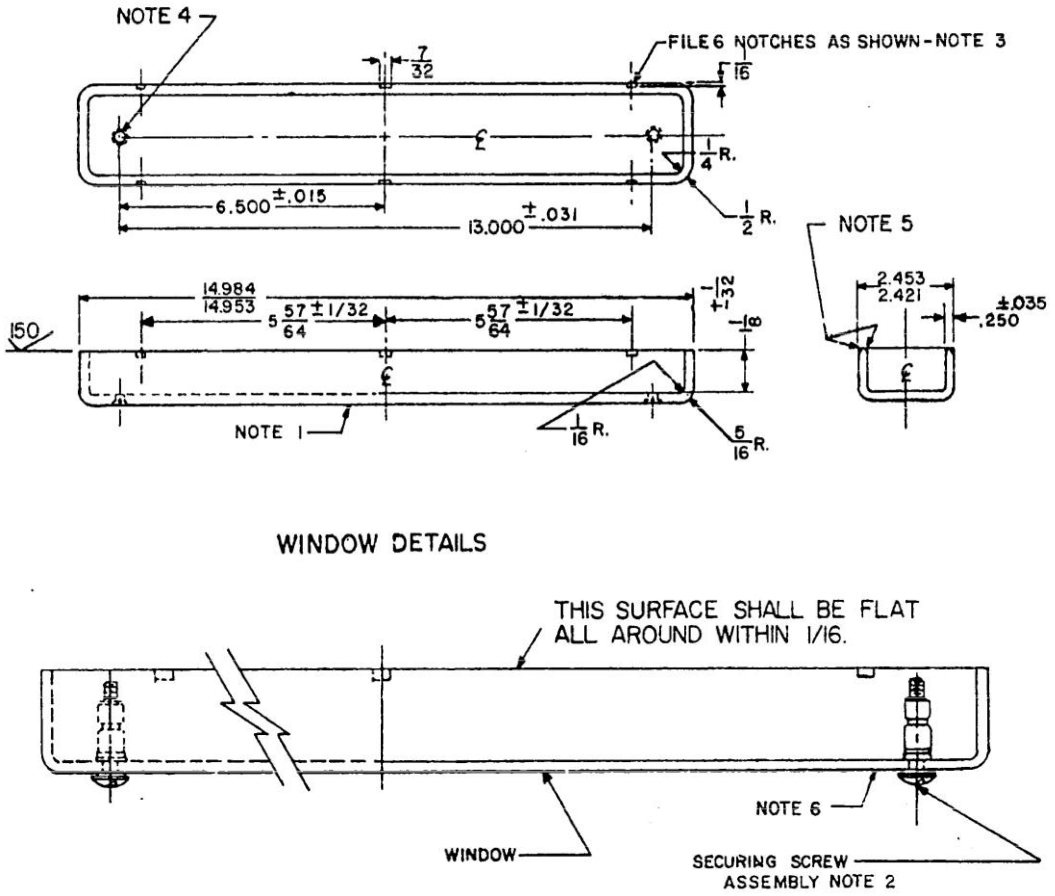


NOTES:

1. Material shall be ¼ inch thick, clear acrylic plastic, Type CLEAR-1 of MIL-DTL-24191.
2. Two long window securing screw assemblies shall be installed in accordance with 3.7.6.3 of MIL-DTL-16377 except that the louver frame shall be inserted between leather washer and captive bushing.
3. Notches shall be filed as shown in accordance with 3.7.6.4 of MIL-DTL-16377.
4. Drill 0.191-inch diameter hole and counterbore opposite side 0.323-inch diameter, 0.050-inch deep. Break sharp edges on the inside surface on all holes.
5. Break sharp edges 1/64 inch around entire window.
6. The inside surface shall be finished white as shown in accordance with 3.8.4.1 of MIL-DTL-16377.
7. When required, one Type VI label shall be applied on the outside surface in accordance with 3.9.1.8.6 of MIL-DTL-16377.
8. The louver assembly shall be finished white in accordance with 3.8.4.2 of MIL-DTL-16377.

FIGURE 6. Window assembly, clear and white, directional, for symbols 79.1 and 79.2.

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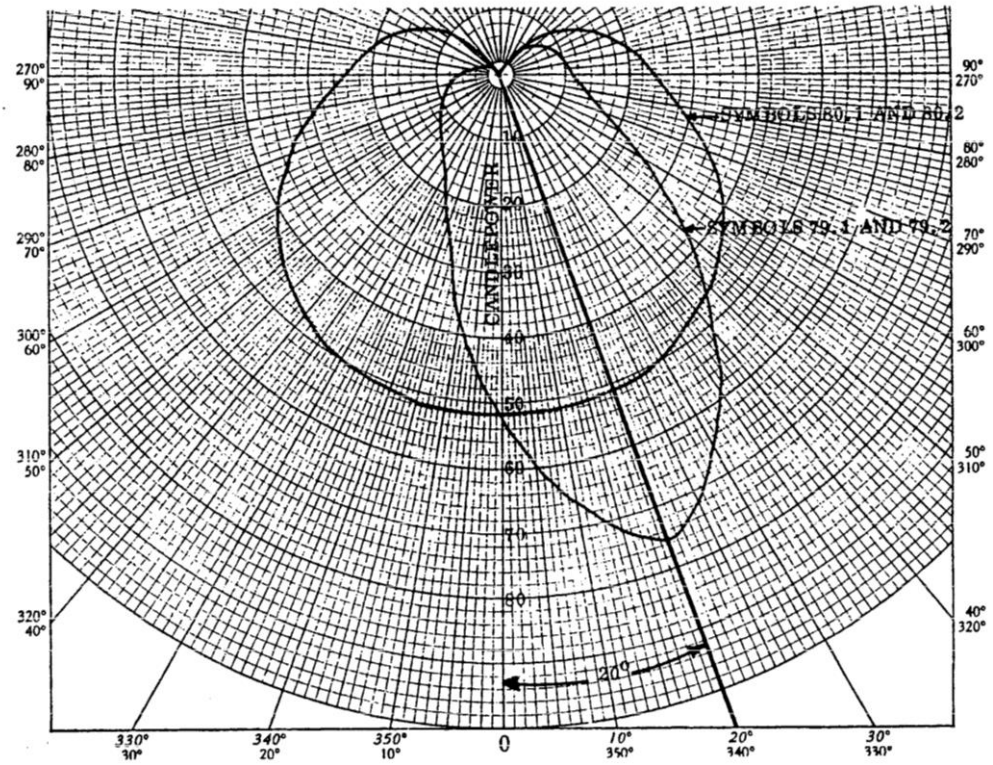


NOTES:

1. Material shall be ¼ inch thick, clear acrylic plastic, Type CLEAR-1 of MIL-DTL-24191.
2. Two long window securing screw assemblies shall be installed in accordance with 3.7.6.3 of MIL-DTL-16377 except that the louver frame shall be inserted between leather washer and captive bushing.
3. Notches shall be filed as shown in accordance with 3.7.6.4 of MIL-DTL-16377.
4. Drill 0.191 diameter hole and counterbore opposite side 0.323 diameter, 0.050 deep. Break sharp edges on the inside surface on all holes.
5. Break sharp edges 1/64 inch around entire window.
6. When required, one Type VI label shall be applied on the outside surface in accordance with 3.9.1.8.6 of MIL-DTL-16377.

FIGURE 7. Window assembly, white, diffusing, for symbols 80.1 and 80.2.

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*ANGLE	SYMBOL NUMBER	
	79.1 and 79.2	80.1 and 80.2
0	76	53
5	72	52
10	66	52
15	60	52
20	52	51
25	45	51
30	38	50
35	29	49
40	26	47
45	20	46
50	18	44
55	15	42
60	13	40

*ANGLE	SYMBOL NUMBER	
	79.1 and 79.2	80.1 and 80.2
65	12	37
70	11	35
75	11	31
80	11	29
85	10	25
90	10	24
95	9	22
100	9	21
105	8	20
110	6	17
115	6	16
120	5	14
125	4	11

* Measured at an angle of 20° from vertical for Symbols 79.1 and 79.2 only.

PHOTOMETRIC TEST DATA:

Distance:	10 feet.
Plane:	At right angles to long axis of fixture through geometric center of lamp.
Voltage:	115 volts, 60 hertz.
Ambient temperature:	76 °F (24.4 °C).
Lamps:	Use lamps furnished with fixtures seasoned for at least 100 hours.
Ballast:	Use ballast furnished with fixture.
Lamp correction:	Distribution curves based on 380 lumens per lamp.

FIGURE 8. Candlepower distribution curves.

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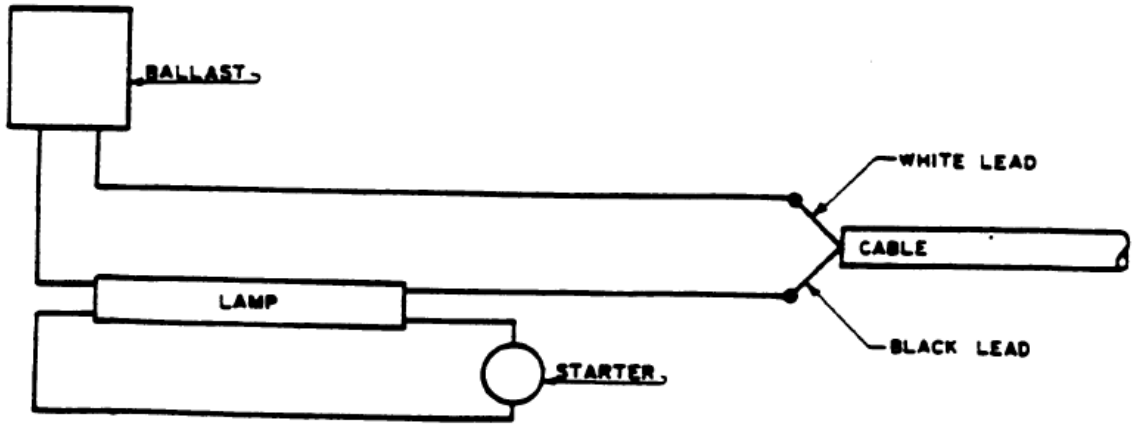


FIGURE 9. Wiring diagram.

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FLUORESCENT REQUIREMENTS:

CLASSIFICATION:

Type: 1
Class: 2

SYMBOL NUMBERS:

79.1: Complete lighting fixture assembly, standard mounting, with one 8-watt cool white fluorescent lamp and clear and white window assembly, directional.

79.2: Complete lighting fixture assembly, flush mounting, with one 8-watt cool white fluorescent lamp and clear and white window assembly, directional.

80.1: Complete lighting fixture assembly, standard mounting, with one 8-watt cool white fluorescent lamp and white diffusing window assembly.

80.2: Complete lighting fixture assembly, flush mounting, with one 8-watt cool white fluorescent lamp and white diffusing window assembly.

DESIGN:

Electrical rating:	8 watts, 120 volts, 60 hertz.
Dimensions and configurations:	See figures 1 through 5 and figure 10 of MIL-DTL-16377.
Photometric data:	See figure 8.
Lampholder spacing:	See figure 21 of MIL-DTL-16377.
Wiring diagram:	See figure 22 of MIL-DTL-16377.
Mounting dimensions:	
Symbols 79.1 and 80.1:	See figure 25 of MIL-DTL-16377.
Symbols 79.2 and 80.2:	See figure 26 of MIL-DTL-16377.
Window assembly:	
Symbols 79.1 and 79.2:	See figure 6.
Symbols 80.1 and 80.2:	See figure 7.
Enclosure:	Watertight. See 3.5.11 of MIL-DTL-16377. The enclosure shall be made of 3/32-inch minimum thickness cast aluminum.
Cable:	10 feet of Type LSDHOF-3 in accordance with MIL-DTL-24643/3.
Weight:	
Symbols 79.1 and 80.1:	4 pounds 3 ounces maximum.
Symbols 79.2 and 80.2:	5 pounds 2 ounces maximum.
Designation and marking:	The following labels shall be furnished in accordance with MIL-DTL-16377 applied as shown on figures 1 and 2:

a. Two Type I.

b. One Type IV for Symbols 79.1 and 79.2 only. Apply one Type IV label as shown on figure 2 bearing legend "THIS SIDE FACES AREA TO BE LIGHTED."

c. One Type IV. The information to be shown on the Type IV label shall be "USE THE FOLLOWING REPAIR PARTS ONLY:" followed by a list of all standard parts shown in table I and window assemblies of table V by name and stock number only.

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SSL REQUIREMENTS:

CLASSIFICATION:

Type: III
Class: 2

SYMBOL NUMBER:

79.1SSL: Complete lighting fixture assembly, standard mounting, with LED luminaire and white window assembly.
79.2SSL: Complete lighting fixture assembly, flush mounting, with LED luminaire and white window assembly.
80.1SSL: Complete lighting fixture assembly, standard mounting, with LED luminaire and white diffusing window assembly.
80.2SSL: Complete lighting fixture assembly, flush mounting, with LED luminaire and white diffusing window assembly.

DESIGN:

Electrical rating:	8 watts maximum, 120 volts, 60 hertz.
Dimensions and configuration:	See figures 1 through 5 and figure 10 of MIL-DTL-16377.
Photometric data:	See figure 8 and 3.7.1 of MIL-DTL-16377.
Chromaticity:	See 3.7.4 of MIL-DTL-16377.
Fail-safe circuit:	See 3.17 of MIL-DTL-16377.
Wiring:	See figure 22 of MIL-DTL-16377.
Mounting dimensions:	
Symbols 79.1SSL and 80.1SSL:	See figure 25 of MIL-DTL-16377.
Symbols 79.2SSL and 80.2SSL:	See figure 26 of MIL-DTL-16377.
Window assembly:	
Symbols 79.1SSL and 79.2SSL:	See figure 6.
Symbols 80.1SSL and 80.2SSL:	See figure 7.
Enclosure:	Watertight. See 3.5.11 of MIL-DTL-16377. The enclosure shall be made of 3/32-inch minimum thickness cast aluminum.
Cable:	10 feet of Type LSDHOF-3 in accordance with MIL-DTL-24643/3.
Weight:	
Symbols 79.1SSL and 80.1SSL:	4 pounds 3 ounces maximum.
Symbols 79.2SSL and 80.2SSL:	5 pounds 2 ounces maximum.
Designation and marking:	The following labels shall be furnished in accordance with MIL-DTL-16377 applied as shown on figures 1 and 2:

a. Two Type I.
b. One Type IV for Symbols 79.1SSL and 79.2SSL only. Apply one Type IV label as shown on figure 2 bearing legend “THIS SIDE FACES AREA TO BE LIGHTED.”
c. One Type IV. The information to be shown on the Type IV label shall be “USE THE FOLLOWING REPAIR PARTS ONLY:” followed by:
   (1) A list of all standard parts shown in table I and window assemblies of table V by name and stock number only, if appropriate, or
   (2) A list of available replacement parts and stock numbers, or if unavailable, manufacturer’s part numbers.

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EXAMINATION AND TESTS:

Examinations and tests shall be as specified in MIL-DTL-16377 and the following:

First article inspection:
Complete fixtures: See table II for Type I and table III for Type III.
Window assemblies: See table IV.
Conformance:
Comparison inspection: Same as first article inspection.
Conformance inspection: See MIL-DTL-16377.

GENERAL INFORMATION:

Assignment of military part numbers and general information is shown in table V.

TABLE I. Standard parts.

Name	Specification or Military Part Number	National Stock Number (NSN)	Remarks
Lamp	ANSI/ANSLG C78.81	6240-00-299-5546	Industry number F8T5/CW
Lamplock	M16377/40-001	6250-00-399-7039	
Lampholder	UL 496	6250-00-377-8590	SUPP. SA, FIG. SA2.1
Startersocket	UL 496	6250-00-295-2738	SUPP. SA, FIG. SA2.4
Starter	NEMA ANSI C78.180	6250-00-299-5962	Commercial designation: FS5
Ballast	M16377/45-001	6250-00-295-2735	
Shock mount	M16377/41-010	6210-00-973-1748	See 3.5.16 of MIL-DTL-16377
Broad band blue filter tube	M16377/42-002	6210-00-503-2810	Not furnished, see 3.7.9 of MIL-DTL-16377
Gray filter tube (2.5% transmission)	M16377/42-014	6210-01-313-8170	Not furnished, see 3.7.9 of MIL-DTL-16377
Gray filter tube (1% transmission)	M16377/42-013	6210-01-351-3943	Not furnished, see 3.7.9 of MIL-DTL-16377
Red filter tube	M16377/42-001	6210-00-399-6297	Not furnished, see 3.7.9 of MIL-DTL-16377

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TABLE II. First article and comparison inspection for complete type I fixtures. [1/]

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Noise	3.5.14	4.8.9	
Continuity of grounding	3.6.13.4	4.8.12	
Optical uniformity	3.7.6.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Light output	3.7.1	4.8.6.1	Candlepower
Magnetic permeability	3.4.8	4.8.18	

NOTE:
[1/] Requirement and test method paragraph numbers refer to MIL-DTL-16377.

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TABLE III. First article and comparison inspection for complete type III fixtures. [1/]

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Power interface	3.6.18	4.8.21	
Power quality	3.6.18.1	4.8.22	
Voltage transient spike	3.6.18.2	4.8.23	
Electromagnetic interference	3.6.14	4.8.16	
Dielectric withstanding voltage	3.6.15	4.8.2	[2/]
Insulation resistance	3.6.16	4.8.3	[2/]
Enclosure effectiveness	3.5.11	4.8.14.4	[2/] Watertight
Shock	3.5.12	4.8.7	[2/]
Vibration	3.5.13	4.8.8	[2/]
Enclosure effectiveness	3.5.11	4.8.14.4	[2/]Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	[2/]
Insulation resistance	3.6.16	4.8.3	[2/]
Noise	3.5.14	4.8.9	[2/]
Continuity of grounding	3.6.13.4	4.8.12	[2/]
Photometrics	3.7.4	4.8.6	
Brightness	3.7.4.1	4.8.6.4	
Chromaticity	3.7.4	4.8.6.5	
Correlated color temperature (CCT), white light only	3.7.4.1.1	4.8.6.6	
Lumen maintenance	3.7.11	4.8.20	
Fail-safe circuit design	3.17	4.8.24	
Optical uniformity	3.7.6.1.1	4.8.4	
Stress relief	3.8.5	4.8.5	
Magnetic permeability	3.4.8	4.8.18	

NOTES:

[1/] Requirement and test method paragraph numbers refer to MIL-DTL-16377.

[2/] In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

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TABLE IV. First article and comparison inspection for window assemblies. [1/]

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Optical uniformity	3.7.6.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Light output	3.7.1	4.8.6.1	Candlepower

NOTE:
[1/] Requirement and test method paragraph numbers refer to MIL-DTL-16377.

TABLE V. Part number and general information.

Military Part Number M16377/14	Item Name	National Stock Number (NSN)	Formerly Covered By
79.1	Complete fixture, Symbol 79.1	6210-00-578-7703	MS16673-79
79.2	Complete fixture, Symbol 79.2	6210-00-519-5792	
80.1	Complete fixture, Symbol 80.1	6210-00-892-1715	MS16673-80
80.2	Complete fixture, Symbol 80.2	6210-01-088-8990	
79.1SSL	Complete fixture, solid state, Symbol 79.1SSL		
79.2SSL	Complete fixture, solid state, Symbol 79.2SSL		
80.1SSL	Complete fixture, solid state, Symbol 80.1SSL		
80.2SSL	Complete fixture, solid state, Symbol 80.2SSL		
001	Window assembly, directional	6210-00-553-6793	MS16891-61
002	Window assembly, white diffusing	6210-00-647-2276	MS16897-66

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Navy – SH

Review activity:
DLA – IS

Preparing activity:
Navy – SH
(Project 6210-2012-028)

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